

# Random Network Routing Evaluation

Autonomous Networks Research Group (ANRG)  
University of Southern California

## Contents

|          |                                                           |          |
|----------|-----------------------------------------------------------|----------|
| <b>1</b> | <b>Evaluation Setup</b>                                   | <b>2</b> |
| 1.1      | Topology Statistics per Density Level . . . . .           | 2        |
| <b>2</b> | <b>Makespan vs. Density</b>                               | <b>2</b> |
| <b>3</b> | <b>Best Routing Scheme per Density and Scheduler</b>      | <b>3</b> |
| 3.1      | Small DAG . . . . .                                       | 3        |
| 3.2      | Large DAG (30 tasks) . . . . .                            | 4        |
| <b>4</b> | <b>Full Results Tables</b>                                | <b>4</b> |
| 4.1      | L150: $L = 150$ m, avg. degree 24.0 . . . . .             | 4        |
| 4.2      | L200: $L = 200$ m, avg. degree 17.1 . . . . .             | 5        |
| 4.3      | L250: $L = 250$ m, avg. degree 12.4 . . . . .             | 5        |
| 4.4      | L300: $L = 300$ m, avg. degree 8.7 . . . . .              | 6        |
| 4.5      | L350: $L = 350$ m, avg. degree 6.7 . . . . .              | 6        |
| 4.6      | L400: $L = 400$ m, avg. degree 5.8 . . . . .              | 7        |
| 4.7      | L500: $L = 500$ m, avg. degree 3.7 . . . . .              | 7        |
| <b>5</b> | <b>Analysis</b>                                           | <b>8</b> |
| 5.1      | HEFT-1 Dominance Is Robust Across All Densities . . . . . | 8        |
| 5.2      | SH vs. S: A Density-Dependent Reversal . . . . .          | 8        |
| 5.3      | Best Routing Transitions with Density . . . . .           | 8        |
| 5.4      | Random vs. Grid Topology . . . . .                        | 9        |
| <b>6</b> | <b>Reproducing These Results</b>                          | <b>9</b> |

# 1 Evaluation Setup

Does the relative performance ranking of HEFT, HEFT-1, and HEFT-2 hold on random graphs? We place **50 nodes** uniformly at random in an  $L \times L$  square and create bidirectional links between all pairs within  $R = 80$  m. Varying  $L$  controls network density. We run the same DAGs, routing schemes, and interference model as the grid evaluation.

| Parameter           | Value                                                     |
|---------------------|-----------------------------------------------------------|
| Nodes               | 50, positions uniform random in $[0, L]^2$                |
| Communication range | $R = 80$ m (csma_bianchi PHY rate from distance)          |
| Connectivity        | guaranteed by minimum spanning bridge if disconnected     |
| Topology seed       | 42 (fixed per density level)                              |
| Seeds per combo     | 30 (simulation seeds 1–30)                                |
| Schedulers          | HEFT (calibrated), HEFT-1, HEFT-2                         |
| Routing schemes     | W, S, SH, GS, GO, GSD, GSD-D                              |
| DAGs                | Small (8 tasks, fork-join) and Large (30 tasks, pipeline) |
| Interference        | csma_bianchi, 802.11ax 5 GHz 20 MHz, $P_{tx} = 20$ dBm    |

Table 1: Random network evaluation parameters.

## 1.1 Topology Statistics per Density Level

| Level | Side $L$ (m) | Links | Avg. degree |
|-------|--------------|-------|-------------|
| L150  | 150          | 601   | 24.0        |
| L200  | 200          | 427   | 17.1        |
| L250  | 250          | 309   | 12.4        |
| L300  | 300          | 217   | 8.7         |
| L350  | 350          | 168   | 6.7         |
| L400  | 400          | 146   | 5.8         |
| L500  | 500          | 92    | 3.7         |

Table 2: Network topology statistics. All topologies generated with seed 42.

## 2 Makespan vs. Density

Each line shows the mean makespan of the *best routing scheme* for that scheduler at each density level. X-axis is average node degree (higher = denser); Y-axis is mean makespan in seconds (log scale).

| Density | Avg. deg. | HEFT (s) | HEFT-1 (s)    | HEFT-2 (s) | HEFT-1 speedup |
|---------|-----------|----------|---------------|------------|----------------|
| L150    | 24.0      | 31.584   | <b>21.594</b> | 34.633     | 1.5×           |
| L200    | 17.1      | 29.712   | <b>27.861</b> | 35.664     | 1.1×           |
| L250    | 12.4      | 32.688   | <b>23.513</b> | 37.796     | 1.4×           |
| L300    | 8.7       | 31.009   | <b>24.388</b> | 28.641     | 1.2×           |
| L350    | 6.7       | 75.816   | <b>24.179</b> | 71.486     | 3.1×           |
| L400    | 5.8       | 45.980   | <b>22.909</b> | 54.861     | 2.0×           |
| L500    | 3.7       | 247.260  | <b>20.926</b> | 174.510    | 11.8×          |

Table 3: Best-routing makespan vs. density — Small DAG (8 tasks, fork-join). Each value is the minimum mean makespan over all routing schemes at that density, averaged over 30 seeds. HEFT-1 speedup =  $\min(\text{HEFT}, \text{HEFT-2}) / \text{HEFT-1}$ .

| Density | Avg. deg. | HEFT (s) | HEFT-1 (s)     | HEFT-2 (s) | HEFT-1 speedup |
|---------|-----------|----------|----------------|------------|----------------|
| L150    | 24.0      | 287.323  | <b>149.899</b> | 308.494    | 1.9×           |
| L200    | 17.1      | 501.174  | <b>391.400</b> | 472.856    | 1.2×           |
| L250    | 12.4      | 1344.548 | <b>449.477</b> | 1336.595   | 3.0×           |
| L300    | 8.7       | 2063.615 | <b>405.276</b> | 2140.430   | 5.1×           |
| L350    | 6.7       | 1939.104 | <b>362.181</b> | 2060.780   | 5.4×           |
| L400    | 5.8       | 2426.675 | <b>927.488</b> | 2388.566   | 2.6×           |
| L500    | 3.7       | 3914.460 | <b>694.194</b> | 5068.567   | 5.6×           |

Table 4: Best-routing makespan vs. density — Large DAG (30 tasks, 5-stage pipeline). Each value is the minimum mean makespan over all routing schemes, averaged over 30 seeds. HEFT-1 speedup =  $\min(\text{HEFT}, \text{HEFT-2}) / \text{HEFT-1}$ .

### 3 Best Routing Scheme per Density and Scheduler

#### 3.1 Small DAG

| Density | Avg. deg. | HEFT (calib.)<br>Route (s) | HEFT-1<br>Route (s) | HEFT-2<br>Route (s) |
|---------|-----------|----------------------------|---------------------|---------------------|
| L150    | 24.0      | S 31.6                     | S 21.6              | S 34.6              |
| L200    | 17.1      | SH 29.7                    | S 27.9              | SH 35.7             |
| L250    | 12.4      | S 32.7                     | S 23.5              | S 37.8              |
| L300    | 8.7       | SH 31.0                    | SH 24.4             | SH 28.6             |
| L350    | 6.7       | SH 75.8                    | SH 24.2             | SH 71.5             |
| L400    | 5.8       | SH 46.0                    | S 22.9              | SH 54.9             |
| L500    | 3.7       | S 247.3                    | S 20.9              | GS 174.5            |

Table 5: Best (routing, makespan) per scheduler for small dag. Makespan in seconds, averaged over 30 seeds.

### 3.2 Large DAG (30 tasks)

| Density | Avg. deg. | HEFT (calib.)<br>Route (s) | HEFT-1<br>Route (s) | HEFT-2<br>Route (s) |
|---------|-----------|----------------------------|---------------------|---------------------|
| L150    | 24.0      | SH 287.3                   | SH 149.9            | SH 308.5            |
| L200    | 17.1      | SH 501.2                   | SH 391.4            | SH 472.9            |
| L250    | 12.4      | SH 1344.5                  | S 449.5             | S 1336.6            |
| L300    | 8.7       | S 2063.6                   | SH 405.3            | SH 2140.4           |
| L350    | 6.7       | S 1939.1                   | SH 362.2            | S 2060.8            |
| L400    | 5.8       | SH 2426.7                  | W 927.5             | SH 2388.6           |
| L500    | 3.7       | S 3914.5                   | GSD-D 694.2         | GSD-D 5068.6        |

Table 6: Best (routing, makespan) per scheduler for large dag (30 tasks). Makespan in seconds, averaged over 30 seeds.

## 4 Full Results Tables

Mean makespan (seconds) averaged over 30 seeds. **Bold green**: overall best for that density+DAG. **Red**: worst in that scheduler column.

### 4.1 L150: $L = 150$ m, avg. degree 24.0

#### Small DAG

| Routing | HEFT (s)       | HEFT-1 (s)     | HEFT-2 (s)     |
|---------|----------------|----------------|----------------|
| W       | 494.056        | 43.651         | 520.163        |
| S       | 31.584         | <b>21.594</b>  | 34.633         |
| SH      | 36.964         | 21.956         | 35.639         |
| GO      | 536.541        | 242.145        | 612.818        |
| GS      | 589.876        | <b>286.759</b> | 562.633        |
| GSD     | <b>640.136</b> | <b>286.083</b> | <b>639.593</b> |
| GSD-D   | 414.070        | 144.089        | 409.654        |

#### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 1530.829        | 542.056         | 1504.689        |
| S       | 715.431         | 198.784         | 675.535         |
| SH      | 287.323         | <b>149.899</b>  | 308.494         |
| GO      | 3540.427        | 2117.429        | 3383.308        |
| GS      | <b>3567.395</b> | <b>2501.810</b> | <b>3589.212</b> |
| GSD     | 3256.272        | 2192.774        | 3309.441        |
| GSD-D   | 2601.980        | 1805.307        | 2568.484        |

## 4.2 L200: $L = 200$ m, avg. degree 17.1

### Small DAG

| Routing | HEFT (s)        | HEFT-1 (s)     | HEFT-2 (s)      |
|---------|-----------------|----------------|-----------------|
| W       | 517.000         | 43.892         | 460.683         |
| S       | 38.313          | <b>27.861</b>  | 38.134          |
| SH      | 29.712          | 28.909         | 35.664          |
| GO      | 872.239         | 411.368        | <b>1184.579</b> |
| GS      | <b>1088.416</b> | <b>489.974</b> | 888.119         |
| GSD     | 990.931         | 483.804        | 936.577         |
| GSD-D   | 459.645         | 229.148        | 830.506         |

### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 1943.409        | 1079.533        | 1822.756        |
| S       | 831.126         | 410.010         | 860.765         |
| SH      | 501.174         | <b>391.400</b>  | 472.856         |
| GO      | <b>4049.688</b> | 3425.020        | <b>4131.985</b> |
| GS      | 3973.898        | <b>4263.847</b> | 3671.203        |
| GSD     | 3531.402        | 3902.399        | 3652.316        |
| GSD-D   | 3131.900        | 2520.026        | 2981.828        |

## 4.3 L250: $L = 250$ m, avg. degree 12.4

### Small DAG

| Routing | HEFT (s)       | HEFT-1 (s)     | HEFT-2 (s)     |
|---------|----------------|----------------|----------------|
| W       | <b>609.587</b> | 29.521         | 546.569        |
| S       | 32.688         | <b>23.513</b>  | 37.796         |
| SH      | 41.687         | 25.843         | 38.534         |
| GO      | 596.181        | 106.264        | 405.795        |
| GS      | 250.967        | 91.428         | 314.147        |
| GSD     | 468.916        | <b>118.211</b> | <b>629.508</b> |
| GSD-D   | 190.333        | 84.703         | 209.977        |

### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 1812.396        | 626.690         | 1852.661        |
| S       | 1357.554        | <b>449.477</b>  | 1336.595        |
| SH      | 1344.548        | 559.928         | 1361.136        |
| GO      | <b>3063.776</b> | 1262.239        | 2986.991        |
| GS      | <b>3048.733</b> | 1820.238        | <b>3023.702</b> |
| GSD     | 2530.877        | <b>1835.618</b> | 2602.391        |
| GSD-D   | 1951.639        | 1226.291        | 1924.656        |

#### 4.4 L300: $L = 300$ m, avg. degree 8.7

##### Small DAG

| Routing | HEFT (s)        | HEFT-1 (s)     | HEFT-2 (s)     |
|---------|-----------------|----------------|----------------|
| W       | 847.135         | 28.512         | 720.078        |
| S       | 34.708          | 26.982         | 33.490         |
| SH      | 31.009          | <b>24.388</b>  | 28.641         |
| GO      | <b>1067.400</b> | <b>151.303</b> | 730.414        |
| GS      | 916.827         | 129.153        | 576.451        |
| GSD     | 361.737         | 109.158        | <b>871.740</b> |
| GSD-D   | 476.092         | 116.086        | 475.722        |

##### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 2639.992        | 809.067         | 2726.267        |
| S       | 2063.615        | 428.149         | 2156.510        |
| SH      | 2233.182        | <b>405.276</b>  | 2140.430        |
| GO      | <b>2810.638</b> | 1078.668        | <b>2859.602</b> |
| GS      | <b>2822.218</b> | 1027.168        | 2807.280        |
| GSD     | 2769.135        | <b>1471.472</b> | 2725.680        |
| GSD-D   | 2740.669        | 998.452         | 2500.433        |

#### 4.5 L350: $L = 350$ m, avg. degree 6.7

##### Small DAG

| Routing | HEFT (s)        | HEFT-1 (s)    | HEFT-2 (s)      |
|---------|-----------------|---------------|-----------------|
| W       | 692.048         | 24.745        | 985.791         |
| S       | 414.203         | 25.020        | 331.441         |
| SH      | 75.816          | <b>24.179</b> | 71.486          |
| GO      | <b>1115.620</b> | 40.942        | <b>1312.777</b> |
| GS      | 1015.965        | 52.288        | 625.956         |
| GSD     | 336.893         | <b>54.115</b> | 1215.382        |
| GSD-D   | <b>1113.997</b> | 39.763        | 917.628         |

##### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 2262.481        | 627.582         | 2378.291        |
| S       | 1939.104        | 767.926         | 2060.780        |
| SH      | 2142.480        | <b>362.181</b>  | 2155.497        |
| GO      | 2577.160        | 880.890         | 2406.193        |
| GS      | 2426.125        | <b>1185.292</b> | <b>2483.959</b> |
| GSD     | <b>2625.803</b> | 1028.508        | 2454.205        |
| GSD-D   | 2575.997        | 891.407         | 2353.488        |

#### 4.6 L400: $L = 400$ m, avg. degree 5.8

##### Small DAG

| Routing | HEFT (s)       | HEFT-1 (s)     | HEFT-2 (s)     |
|---------|----------------|----------------|----------------|
| W       | <b>837.123</b> | 29.321         | <b>689.413</b> |
| S       | 615.175        | <b>22.909</b>  | 615.634        |
| SH      | 45.980         | 23.741         | 54.861         |
| GO      | 503.103        | 207.313        | 512.725        |
| GS      | 440.730        | <b>255.294</b> | 384.038        |
| GSD     | 437.806        | 243.985        | 592.820        |
| GSD-D   | 423.879        | 117.097        | 433.288        |

##### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 2844.226        | <b>927.488</b>  | 2980.537        |
| S       | 2947.258        | 1080.935        | 2919.019        |
| SH      | 2426.675        | 1252.065        | 2388.566        |
| GO      | <b>3331.649</b> | 1229.818        | <b>3336.711</b> |
| GS      | 3154.020        | 1611.463        | 3239.241        |
| GSD     | 3209.028        | <b>1621.605</b> | 3065.852        |
| GSD-D   | 3070.476        | 1202.449        | 3205.536        |

#### 4.7 L500: $L = 500$ m, avg. degree 3.7

##### Small DAG

| Routing | HEFT (s)       | HEFT-1 (s)    | HEFT-2 (s)     |
|---------|----------------|---------------|----------------|
| W       | 356.336        | 21.670        | 210.881        |
| S       | 247.260        | <b>20.926</b> | 392.711        |
| SH      | <b>501.809</b> | 23.151        | <b>465.443</b> |
| GO      | <b>501.795</b> | 22.127        | <b>465.438</b> |
| GS      | 429.050        | 21.384        | 174.510        |
| GSD     | 392.711        | <b>32.377</b> | 356.363        |
| GSD-D   | 318.912        | 21.864        | 318.765        |

##### Large DAG

| Routing | HEFT (s)        | HEFT-1 (s)      | HEFT-2 (s)      |
|---------|-----------------|-----------------|-----------------|
| W       | 5494.190        | 906.925         | 5732.305        |
| S       | 3914.460        | 792.757         | 6166.377        |
| SH      | <b>5924.170</b> | <b>1035.847</b> | <b>6870.982</b> |
| GO      | 4771.168        | 923.603         | 5773.163        |
| GS      | 4544.688        | 860.593         | 5767.751        |
| GSD     | 5560.900        | 938.632         | 5386.756        |
| GSD-D   | 5212.339        | <b>694.194</b>  | 5068.567        |

## 5 Analysis

### 5.1 HEFT-1 Dominance Is Robust Across All Densities

On the regular  $7 \times 7$  grid, HEFT-1 dominated HEFT and HEFT-2 by  $3\text{--}7\times$  because csma.bianchi interference made multi-hop transfers prohibitively expensive, and the uniform grid gave the greedy co-location heuristic many same-node placement options. The random-graph experiments confirm this dominance is not an artifact of the grid structure:

- **Large DAG:** HEFT-1 wins at every density level. At the densest setting (L150, avg. degree 24.0) the best HEFT-1 makespan is 149.9s vs. 287.3s for calibrated HEFT and 308.5s for HEFT-2—a  $1.9\text{--}2.1\times$  advantage. At the sparsest setting (L500, avg. degree 3.7) HEFT-1 still wins: 694.2s vs. 3914.5s for HEFT and 5068.6s for HEFT-2—a  $5.6\text{--}7.3\times$  advantage.
- **Small DAG:** HEFT-1 wins at every density level, with makespans in the range 20–27s regardless of density, while HEFT ranges from 29s to 247s and HEFT-2 from 28s to 174s.
- The HEFT-1 advantage is *larger* at low density. When nodes are sparse, multi-hop paths grow longer and relay interference is more severe; HEFT-1’s co-location bias avoids this cost, whereas calibrated HEFT spreads tasks across nodes forced into long paths.

### 5.2 SH vs. S: A Density-Dependent Reversal

On the 40 m grid evaluation, shortest-path-by-delay (S) outperformed shortest-hop (SH) because diagonal links (56.6m, lower BW) competed with 2-hop cardinal paths (40m, higher BW)—S correctly preferred the higher-bandwidth 2-hop route.

On random graphs the result reverses for the large DAG at high to medium density:

- At L150 (deg 24.0): SH wins for both HEFT and HEFT-2 (HEFT/SH = 287.3s vs. HEFT/S = 292.1s; HEFT-2/SH = 308.5s vs. HEFT-2/S = 328.1s).
- At L200 (deg 17.1): SH wins for all three schedulers. The margin is largest for HEFT-1: HEFT-1/SH = 391.4s vs. HEFT-1/S = 432.2s (11% improvement).
- At L300–L350 (deg 6.7–8.7): SH still wins under HEFT-1 (HEFT-1/SH = 362–405s vs. HEFT-1/S = 429–449s).
- At L250 (deg 12.4) and L400–L500 (deg  $\leq 5.8$ ): S regains the lead for HEFT-1.

The reversal is explained by the nature of random graphs at high density: nodes are placed uniformly at random in a bounded area, so the majority of within-range links have very similar lengths and therefore similar PHY rates. When bandwidth variation across links is small, the key cost of a longer path is the number of relay transmitters it adds to the interference graph. SH’s primary objective of minimising hop count directly minimises that interference cost; S’s delay-minimising objective provides no advantage when all links have similar bandwidth.

### 5.3 Best Routing Transitions with Density

The optimal routing scheme under HEFT-1 shifts as density decreases:

- **Dense (avg. degree  $\geq 17$ ):** SH (large DAG) or S (small DAG). Many short 1-hop paths exist; SH exploits them and avoids relay interference.

- **Medium-dense (avg. degree 8–13):** SH dominates for large DAG; S competitive for small DAG where shorter task chains reduce sensitivity to routing choice.
- **Sparse (avg. degree  $\approx 6$ ):** W (widest-path) wins for large DAG at L400 (HEFT-1/W = 927.5 s). When the network has few alternative paths, maximising the bottleneck bandwidth on the chosen path outweighs hop-count minimisation.
- **Very sparse (avg. degree  $\approx 4$ ):** GSD-D wins for large DAG at L500 (HEFT-1/GSD-D = 694.2 s vs. HEFT-1/W = 906.9 s, HEFT-1/S = 792.8 s). This is the first experiment in either grid or random evaluations in which GSD-D wins under HEFT-1 for a large DAG. At this density, path options are limited but GSD-D’s deferral mechanism avoids the worst interference collisions that occur when all tasks must communicate over the same sparse set of bottleneck links.

## 5.4 Random vs. Grid Topology

Two structural differences explain the divergence from grid results:

1. **Non-uniform node degree.** In a random geometric graph, degree follows a roughly Poisson distribution. Hub nodes (high degree) attract HEFT-1 placements because they can serve multiple tasks locally; leaf nodes (degree 1–2) are effectively unreachable for co-location. On the grid every interior node has degree 8 (with diagonals), so placement options are uniform.
2. **Path length and BW homogeneity.** On the 40 m grid, diagonal links (56.6 m) have meaningfully lower BW than cardinal links (40 m), creating the strong S>SH ordering. On random graphs at high density, inter-node distances within the 80 m comm range are drawn from a near-uniform distribution, so link BWs cluster tightly—making hop count rather than per-link BW the dominant routing cost.

## 6 Reproducing These Results

```
cd ncsim/
python run_random_eval.py
# Output JSON: /tmp/ncsim_random_eval/random_eval_results.json
```